

Design Project: Off-Grid Energy System

1. Project Overview

For your final design project, you will architect a complete, off-grid renewable energy system for a remote residential property, write a report, and present your design to the class.

You must design a robust microgrid that integrates multiple generation sources to reliably meet the electrical, domestic hot water (DHW), and space heating/cooling demands of the site. You are expected to optimize the system for both thermodynamic efficiency and economic viability.

Team Formation: You will be randomly assigned to engineering design teams of three. You must collaborate to divide the analysis, modeling, and reporting responsibilities.

2. Site Specifications

Your system will be deployed on a coastal property with the following characteristics:

- **Location:** 10-acre forested parcel situated just inland from Yachats, on the Oregon Coast.
- **Structure:** A small house of cabin with a $UA = 0.16 \text{ kW/degK}$.
 $\text{Steady State Heater Power} = UA * (T_{\text{indoor}} - T_{\text{outdoor}})$
- **Topography & Wind:** The parcel is heavily wooded, but the property boundaries and local zoning permit the clearing of up to 1.5 acres of timber to establish a residential footprint and an unobstructed wind corridor for a small-scale wind turbine.
- **Hydrology:** A year-round stream intersects the property boundary. Topographical surveys indicate a usable head of **35 feet**, with a minimum guaranteed seasonal flow rate of **1.8 cubic feet per second (cfs)**.

3. Provided Data

You will be provided with a `Yachats_2024.csv` file via the [ESE 450 GitHub page](#). This file contains a full year of hourly (8,760 hours) time-series data for 2024, including:

- Hourly electrical load (kW) by category
- Domestic hot water demand (Gallons/hr at 120°F)
- Ambient temperature
- Wind speed

4. Required Deliverables & Tasks

A. Resource Assessment

Use mainstream computational tools to assess energy resources on the site.

B. System Architecture & Sizing

Design a hybrid energy system capable of satisfying the demands in the provided CSV. Your design must specify:

- **Generation:** Sizing and specifications for the Solar PV array, Wind Turbine, and Micro-Hydro generator.
- **Storage & Dispatch:** Battery bank chemistry, capacity, and charge/discharge limits.
- **Load Shedding:** Equipment or system selected for shedding electrical loads.
- **Electric Vehicles:** Is an EV part of the system?
- **Control Law:** The controls that will decide when and at what rate to generate power, dispatch energy, store or dispatch from energy storage systems, and when to shed loads and which ones to shed. This should be implemented as a Python function that returns a control vector with one element for each controllable device in the system and should be tested within your simulation environment.
- **Thermal Systems:** Equipment selected to meet the HVAC and DHW loads (e.g., air-to-water heat pumps, resistive elements, thermal storage tanks).
- **Balance of System (BOS):** Inverters, charge controllers, and dump loads (important for hydro and wind over-generation).

C. Simulation

Build a simulation in Python that will demonstrate hour-by-hour operation of your system during the 2024 calendar year. You will write a control law function and a state update function. The control law will determine the power setpoints and generation levels for each controllable device in the system and output the controls as a vector. The state update function will update the state of any devices and environmental conditions such as house temperature given their previous state and given the control vector. Simulation outputs should be time-series of quantities such as:

- Power of each generator and how much each device is being curtailed
- Power consumption of each category of electrical load and any active load shedding.
- Power or fuel consumption of non-electrical loads

- Energy storage device states (e.g. state of charge, and charge/discharge rate)
- Temperatures of hot water and home interior temperatures

D. Economic Analysis

You must calculate the expected annual cost of your system using standard LCOE-type methodology.

5. Grading & Evaluation Criteria

Your team's final report and presentation will be evaluated based on the following rubric:

- **System Reliability & Load Satisfaction (30%):** Does the modeled system meet the 8,760-hour demand profile? Are instances of unmet load (load shedding and blackouts) properly minimized and quantified?
- **Economic Optimization (30%):** Is the LCOE minimized without sacrificing reliability? Are the CAPEX and OPEX assumptions realistic and properly cited?
- **System Resilience & Redundancy (20%):** How well does the system handle extreme weather anomalies common to the Oregon Coast (e.g., prolonged winter cloud cover blocking solar, or freezing temps affecting hydro flow)? What is the plan in case of an electromagnetic pulse (EMP) takes out all devices utilizing integrated circuits?
- **Presentation and Engineering Documentation (20%):** Clarity of the final report, accuracy of thermodynamic calculations, and professionalism of the system schematics.

Submission Format: Submit a comprehensive engineering report (PDF format, max 20 pages excluding appendices) and your supporting computational models (Excel, Python scripts, etc.) by the date specified on the syllabus. Presentations will be approximately